Pituitary/DI/SIADH — Diabetes insipidus causes (central vs nephrogenic)

1. AVP deficiency wing (central)

WHAT YOU SEE

Left wing labeled AVP deficiency, formerly central DI

WHAT IT ENCODES

Central DI (new name: AVP DEFICIENCY) is failure of the hypothalamus/posterior pituitary to synthesize or release ADH, so urine cannot be concentrated. Labs show dilute urine (Uosm low) with rising serum Na/osm; the defining feature is that it RESPONDS to desmopressin — urine osmolality climbs >50% after DDAVP. Posterior-pituitary bright spot is typically absent on MRI.

BOARD TRAP

WRONG: central DI is renal resistance to ADH. Discriminator — central DI is a PRODUCTION/secretion deficit (low endogenous AVP, low copeptin) that corrects with exogenous DDAVP; renal resistance with high endogenous AVP is nephrogenic DI.

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2. AVP resistance wing (nephrogenic)

WHAT YOU SEE

Center wing labeled AVP resistance, formerly nephrogenic DI

WHAT IT ENCODES

Nephrogenic DI (new name: AVP RESISTANCE) is failure of the collecting duct to respond to ADH despite normal or HIGH circulating AVP (and high baseline copeptin >21.4 pmol/L). The defect is at the V2 receptor or downstream AQP2. Hallmark: little or NO rise in urine osmolality (<50%) after a desmopressin challenge.

BOARD TRAP

WRONG: nephrogenic DI is due to low ADH production. Discriminator — AVP production is intact (often elevated) in nephrogenic DI; the problem is end-organ resistance, so DDAVP fails to concentrate the urine — the opposite of central DI.

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3. Primary polydipsia wing

WHAT YOU SEE

Right wing labeled primary polydipsia

WHAT IT ENCODES

Primary polydipsia is excessive water intake that APPROPRIATELY suppresses AVP, so it is not true DI. Chronic overdrinking washes out the medullary concentration gradient, so maximal urine concentrating ability is blunted and can mimic DI. Baseline serum Na/osm is LOW-normal (the patient is overloaded with water), the opposite of true DI.

BOARD TRAP

WRONG: primary polydipsia is treated with desmopressin. Discriminator — giving DDAVP to someone who keeps drinking causes dangerous water intoxication/hyponatremia; treatment is fluid restriction and behavioral measures. Key clue: low baseline serum osm in polydipsia vs high in true DI.

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4. Trauma / surgery

WHAT YOU SEE

Surgeon portrait, trauma

WHAT IT ENCODES

Central DI causes — head trauma and pituitary/hypothalamic surgery (e.g., transsphenoidal adenoma resection). Post-surgical/post-traumatic central DI is classically TRANSIENT or follows a triphasic pattern: initial DI (axon shock) → transient SIADH (release of stored AVP from dying neurons) → permanent DI if enough neurons are destroyed.

BOARD TRAP

WRONG: head trauma causes nephrogenic DI. Discriminator — trauma/surgery damages the hypothalamic-stalk-posterior pituitary axis, producing AVP-deficient CENTRAL DI that responds to DDAVP; nephrogenic causes are renal/drug/metabolic.

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5. Tumor

WHAT YOU SEE

Brain tumor portrait

WHAT IT ENCODES

Central DI causes — mass lesions of the hypothalamic-pituitary region: craniopharyngioma, germinoma, pituitary macroadenoma compressing the stalk, and metastases (classically breast and lung). Suprasellar/stalk tumors are more likely to cause DI than purely intrasellar ones.

BOARD TRAP

WRONG: pituitary/hypothalamic tumors affect only the anterior pituitary. Discriminator — stalk and suprasellar lesions readily disrupt the posterior pituitary/hypothalamic AVP neurons, producing central DI; new-onset polyuria after a sellar mass should prompt a DI workup.

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6. Infiltrative / infectious

WHAT YOU SEE

Infectious + sarcoid/histiocytosis portraits

WHAT IT ENCODES

Central DI causes — infiltrative and inflammatory disease: sarcoidosis, Langerhans cell histiocytosis, granulomatosis with polyangiitis, IgG4 disease; infections (meningitis, encephalitis, TB); and lymphocytic (autoimmune) hypophysitis. These often produce stalk thickening on MRI.

BOARD TRAP

WRONG: sarcoidosis causes nephrogenic DI. Discriminator — sarcoid/histiocytosis infiltrate the hypothalamus/stalk and destroy AVP-producing neurons → CENTRAL DI (DDAVP-responsive). Nephrogenic DI's metabolic culprits are lithium, hypercalcemia, and hypokalemia.

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7. Idiopathic / autoimmune

WHAT YOU SEE

A blank-faced framed portrait in the AVP-deficiency wing labelled IDIOPATHIC — the featureless face means no visible structural lesion (often autoimmune anti-AVP-cell antibodies).

WHAT IT ENCODES

A large fraction (~30–50%) of central DI is idiopathic, many of these autoimmune (anti-AVP-cell antibodies). Familial central DI also exists (autosomal-dominant mutation in the AVP-neurophysin II gene on chromosome 20), as does Wolfram/DIDMOAD syndrome (DI, DM, optic atrophy, deafness).

BOARD TRAP

WRONG: central DI is always due to a visible structural lesion. Discriminator — a substantial share is idiopathic/autoimmune or genetic with a normal-appearing pituitary; absence of a mass does not exclude central DI, so confirm with copeptin/DDAVP testing.

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8. Lithium (most common acquired)

WHAT YOU SEE

Big LITHIUM bottle, labeled most common acquired cause

WHAT IT ENCODES

Lithium is the MOST COMMON acquired cause of nephrogenic DI. It enters collecting-duct principal cells through the apical ENaC channel and downregulates AQP2 expression, impairing the water-reabsorption response to AVP. ~20–40% of chronic lithium users develop some concentrating defect; effect may persist even after stopping the drug. Amiloride (ENaC blocker) is first-line because it blocks lithium entry.

BOARD TRAP

WRONG: demeclocycline is the most common cause of nephrogenic DI. Discriminator — lithium is by far the leading acquired cause; demeclocycline also causes nephrogenic DI but is uncommon and is in fact used THERAPEUTICALLY in SIADH to induce a controlled nephrogenic DI.

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9. Hypercalcemia

WHAT YOU SEE

A framed panel in the AVP-resistance wing labelled METABOLIC with a high-calcium cue — hypercalcemia activating the CaSR and downregulating AQP2 → renal resistance.

WHAT IT ENCODES

Hypercalcemia (typically Ca >11 mg/dL) causes acquired nephrogenic DI by activating the basolateral calcium-sensing receptor in the thick ascending limb/collecting duct, which downregulates AQP2 and impairs the countercurrent gradient. This is reversible when calcium normalizes and is a classic mechanism behind polyuria in hyperparathyroidism or malignancy.

BOARD TRAP

WRONG: hypoCALCEMIA causes nephrogenic DI. Discriminator — it is hyperCALCEMIA (and hypoKALEMIA) that impair urinary concentration; correcting the calcium reverses the polyuria, which is itself a useful diagnostic clue.

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10. Hypokalemia

WHAT YOU SEE

A framed panel in the AVP-resistance wing labelled METABOLIC / low K — chronic hypokalemia wrecking the medullary gradient and AQP2 (mnemonic 'low K, high Ca').

WHAT IT ENCODES

Chronic hypokalemia (K <3 mEq/L) impairs the medullary countercurrent concentrating mechanism and downregulates AQP2, producing reversible nephrogenic DI. Think of it in laxative/diuretic abuse, primary hyperaldosteronism, and prolonged vomiting. Concentrating ability recovers once potassium is repleted.

BOARD TRAP

WRONG: hyperKALEMIA causes nephrogenic DI. Discriminator — the offending electrolyte derangements are hypoKALEMIA and hyperCALCEMIA (mnemonic: 'low K, high Ca' break the kidney's concentrating power); correcting them reverses the DI.

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11. Hereditary nephrogenic

WHAT YOU SEE

Hereditary portraits, kidney/sickle images

WHAT IT ENCODES

Hereditary nephrogenic DI: ~90% is X-linked recessive due to V2 receptor (AVPR2) gene mutations (affects males, presents in infancy with hypernatremic dehydration, failure to thrive, vomiting). The remainder is autosomal (recessive or dominant) from aquaporin-2 (AQP2) gene mutations. Both leave the collecting duct unable to respond to AVP from birth.

BOARD TRAP

WRONG: hereditary nephrogenic DI is caused by an AVP (vasopressin) gene defect. Discriminator — an AVP-gene mutation causes hereditary CENTRAL DI (deficiency); hereditary nephrogenic DI is a RECEPTOR (AVPR2, X-linked) or channel (AQP2) defect — resistance, not deficiency.

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12. Renal disease / sickle cell

WHAT YOU SEE

Kidney + sickle cell images, nephrogenic wing

WHAT IT ENCODES

Other nephrogenic DI causes — sickle cell disease/trait (vaso-occlusion damages the hypoxic renal medulla, blunting concentration even before overt CKD), post-obstructive uropathy (after relief of obstruction), and chronic tubulointerstitial/medullary kidney disease (e.g., medullary cystic disease, amyloid, Sjögren).

BOARD TRAP

WRONG: sickle cell disease causes central DI. Discriminator — sickling damages the renal medulla → impaired concentrating ability = NEPHROGENIC DI (hyposthenuria), with normal/high AVP. Central DI is the hypothalamic-pituitary deficiency state.

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13. Gestational DI

WHAT YOU SEE

A pregnancy/placenta panel labelled GESTATIONAL DI IN PREGNANCY — the placenta secreting vasopressinase that chews up endogenous AVP.

WHAT IT ENCODES

Gestational DI is a transient third-trimester DI caused by placental vasopressinase (cysteine aminopeptidase) degrading endogenous AVP, increasing its clearance 3–4-fold and outpacing supply. It is treated with DESMOPRESSIN because DDAVP is resistant to vasopressinase, and it resolves within weeks postpartum once the placenta is gone.

BOARD TRAP

WRONG: gestational DI is treated with a thiazide. Discriminator — thiazides treat nephrogenic DI; gestational DI is an AVP-degradation problem and responds to vasopressinase-resistant DDAVP. (Note: associate it with pre-eclampsia/HELLP, which raise vasopressinase.)

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14. Polydipsia psychogenic/iatrogenic

WHAT YOU SEE

Right wing: psychiatric + IV-fluid figures

WHAT IT ENCODES

Primary polydipsia subtypes: psychogenic (compulsive water drinking, often in schizophrenia or with anticholinergic dry mouth), dipsogenic (a low/abnormal thirst threshold so AVP is suppressed prematurely), and iatrogenic (over-recommended or excessive prescribed fluids). All feature excess intake with appropriately suppressed AVP.

BOARD TRAP

WRONG: primary polydipsia has HIGH serum osmolality at baseline. Discriminator — overdrinking dilutes the plasma, so baseline serum osm/Na is LOW-normal in primary polydipsia, whereas true DI (water loss) gives high-normal/high serum osm. This single value helps separate the two before formal testing.